A Comparative Study of Temporal and Spatio-Temporal Mortality Forecasting Methods in the United States

Author:
Dániel György Koch, Corvinus University of Budapest and ELTE,
Actuarial and Financial Mathematics Msc 2nd Year
Supervisor:
Péter Vékás, Ph.D, Corvinus University of Budapest

2026

Abstract

The area of mortality forecasting is crucial for actuaries and demographers. In recent times, new efforts have been made to incorporate spatiality into mortality forecasting to further improve forecasting performance and the understanding of mortality dynamics.

This study compares the performance of temporal and spatio-temporal mortality forecasting methods within the Lee–Carter and Li–Lee frameworks to examine whether incorporating spatiality could improve forecasting performance in the United States. It also introduces a Lagged only Spatial Dynamic Panel Linear Model (LSDPLM) estimated by System-GMM, and the estimation of SDPLM with System-GMM in this context. Furthermore, it employs two types of LISA clustering techniques and presents figures showing the distribution of the winning forecasting methods by state to assess whether geographic patterns can be found.

The empirical analysis is conducted on mortality data from the United States, for years 1969 to 2019 and ages 0 to 89 on all states except Alaska and Hawaii using multiple train-test splits for robust performance evaluation.

The results show, that in the United States under the evaluation method applied in this study, the LSDPLM and SDPLM estimated by System-GMM outperform most other methods, with spatio-temporal approaches overall improving prediction accuracy. Additionally, LISA and the winning method distribution figures highlight clusters that provide deeper insight into mortality dynamics in the United States.

These advancements could lead to valuable benefits in areas such as pension, life insurance, and public policymaking.

Contents

1 Introduction **1**

2 Literature review **2**

2.1 Single and multi-population mortality forecasting 2

2.2 Spatio-temporal forecasting methods 4

3 Methods **5**

3.1 Moran's I - Spatial autocorrelation 5

3.2 Local extension of Moran's I - LISA clustering 6

3.3 Spatio-temporal extension of Moran's I 6

3.4 Mortality models 7

3.5 Forecasting methods 8

3.6 Forecasting performance evaluation methodology 12

4 Empirical analysis **12**

4.1 Data sourcing and preprocessing 12

4.2 Analysis workflow 15

4.3 Moran's I results for spatial and spatio-temporal autocorrelation 17

4.4 LISA clustering results 17

4.5 Mortality forecasting 20

4.6 Model and forecasting method evaluation 22

5 Conclusion **29**

References **31**

Appendix **37**

1 Introduction

Mortality forecasting plays a key role in actuarial science and demography by providing estimates of future mortality trends that are essential for understanding how populations will change over time and for anticipating how these changes will affect pension schemes, life insurance policies, and public policymaking, among other potentially affected areas.

The study of mortality has come a long way since Gompertz (1825) introduced his law of mortality. Since then, mortality forecasting has emerged as a field in which, up until the 1980s, methods were deterministic or relied on subjective judgement from experts as described in a review by Booth and Tickle (2008). A crucial step for mortality forecasting was the introduction of a stochastic mortality forecasting model by Lee and Carter (1992) that describes age–time-specific death rates and enables the use of time-series forecasting by having a time-varying mortality index. However, a concern with this model was that it didn't take into account the coherence of the forecasts, and if it were to be applied to multiple populations, forecasts could diverge in an unrealistic way as argued by (Li and Lee, 2005), who proposed a multi-population mortality model taking coherence into account. Besides the many advancements and extensions to single and multi-population models, spatiality and cross-sectional dependency are yet to be thoroughly explored with regard to mortality forecasting. In other fields, models that take spatiality and cross-sectional dependence into account have been in use for some time with great success, as incorporating these factors may improve prediction accuracy of models. For this reason, incorporating them into mortality forecasting seems like an important next step, allowing for not only temporal, but also spatio-temporal forecasts.

In this study, a comparative empirical analysis is conducted of the performance of temporal and spatio-temporal mortality forecasting methods within the Lee–Carter and Li–Lee frameworks, using mortality data from the United States, where such analysis has yet to be done to the best of my knowledge.

The analysis covers the period from 1969 to 2019, ages 0 to 89, and all states except Alaska and Hawaii, with multiple train-test splits for robust forecasting method evaluation.

In addition, it proposes a Lagged-only Spatial Dynamic Panel Model (LSDPLM) estimated by System-GMM, and estimates the Spatial Dynamic Panel Model (SDPLM) by System-GMM alongside previously suggested approaches in this context. Furthermore, two types of LISA clustering techniques are employed to gain further insight into the potential spatial patterns in mortality in the United States. Figures are also presented showing the spatial distribution of the best performing forecasting methods and whether the winning method is spatial or non-spatial by state, which could help identify clusters and additional spatial patterns.

Accordingly, the following research questions are posed:

- Do spatio-temporal mortality forecasting methods outperform temporal methods in the United States under the evaluation method used in this study?
- Does the proposed LSDPLM estimated by System-GMM and the estimation of SDPLM by System-GMM improve forecasting accuracy?
- Are there discernible geographic patterns in mortality in the United States and among the best performing forecasting methods by state?

The remainder of this work is structured as follows: In Chapter 2 a literature review is conducted on single and multi-population forecasting and spatio-temporal forecasting methods, in Chapter 3 the relevant methodology for the study is summarized, in Chapter 4 the LISA clustering and the empirical analysis is conducted and results are presented, and in Chapter 5 a conclusion is written with a discussion of limitations and possible further improvements.

2 Literature review

2.1 Single and multi-population mortality forecasting

The classic model for mortality forecasting was famously developed by Lee and Carter (1992) who introduced a stochastic approach to forecast mortality in a single population via decomposing mortality rates into different components. Their method relies on time-series modelling as the time-dependent element of mortality in their model is reduced to a single index. They demonstrate their model, commonly referred to as the Lee–Carter (LC) model, by forecasting mortality rates in the United States via Random Walk with Drift (RWD).

Since then, many new approaches and advancements have been made that extend their method in various ways with Brouhns et al. (2002) redefining the framework by modelling death counts as variables with a Poisson distribution and using Maximum Likelihood Estimation (MLE), with Currie et al. (2004) introducing the P-spline smoothing method into mortality forecasting which allows for flexible forecasting while avoiding overfitting, and with Cairns et al. (2006) focusing on particularly improving forecasting performance in old age by using the logit of death probabilities when modelling, thus greatly contributing to the field of pension with their Cairns–Blake–Dowd model. Further significant extensions came with the implementation of the Age-Period-Cohort model, which also takes cohort effects into account by Renshaw and Haberman (2006), and its generalized version developed by Hunt and Blake (2014). Though the Age-Period-Cohort model received some criticism for being slow to converge to an optimum solution and for the lack of robustness when it comes to changes in the data by Macdonald et al. (2007); Cairns et al. (2011). These

criticisms were later validated by Hunt and Villegas (2015) who demonstrated the existence of identifiability issue in models that extended the LC model with cohort effects and also made suggestions that improved the speed of convergence and robustness by imposing additional constraints on the model parameters.

Besides the single-population mortality forecasting methods a new approach has also been developed for multi-population forecasting by Li and Lee (2005), motivated by the idea that mortality patterns in closely related populations are likely to be similar in some ways, therefore the differences in the long run are unlikely to increase. In the case of the LC model where this coherence is not taken into account, when applied to multiple similar populations the forecasts can diverge or have unrealistic crossovers, creating unrealistic long run predictions. Meanwhile, this new approach for multi-population forecasting referred to as the Li–Lee (LL) model, allows for coherence by having common parameters describing the whole group and modelling deviations from the common trend for each population with population specific factors. Extensions of the LL method and articles building on the ideas presented by it have also been made over time by Dowd et al. (2011) who proposed a gravity model for joint modelling of mortality in two differently sized populations taking coherence into account based on the LL model, by Wan and Bertschi (2015) who developed a model for coherent forecasting of mortality rates in a small population, more specifically Switzerland, with combining the model introduced by Plat (2009) (that previously took inspiration from Lee and Carter (1992); Renshaw and Haberman (2006); Cairns et al. (2009) to make a model that takes the best parts of the mortality models developed so far, while also trying to eliminate their disadvantages) with concepts from Li and Lee (2005); Jarner and Kryger (2011). Some further extensions among others have been made by Boonen and Li (2017) who extended the LL model by including the real gross domestic product per capita in order to capture the common mortality trend on populations with similar socioeconomic conditions, by Ahmadi and Li (2014) who came up with an approach for achieving coherent multi-population mortality forecasts under the Generalized Linear Modelling (GLM) framework, and by Guibert et al. (2020) who argue that the coherence assumption made by many multi-population models may be too strong of an assumption especially when applied to a large group of different populations and proposes locally coherent forecasts, which enforce coherence only in subgroups of populations.

A recent and rapidly evolving direction in mortality forecasting is the use of deep learning and other machine learning methods, which are particularly well suited for capturing complex non-linear patterns in data. Levantesi and Pizzorusso (2019) uses Decision tree, Random forest and Gradient boosting to improve mortality forecasting accuracy, while Shen et al. (2024) proposed the GT-A (Graph convolutional neural networks and Transformer Network with Adaptive Adjustment Matrix) model to achieve better forecast then traditional models, and Richman and Wüthrich (2021) developed a neural network based extension to multi-population mortality forecasting.

There is a vast amount of further mortality model types with different approaches and modifications besides the ones mentioned here before. Some relevant comprehensive reviews of the literature were written by Janssen (2018); Barrieu et al. (2012); Enchev et al. (2017), and by Booth and Tickle (2008) who reviews mortality forecasting under three approaches: expectation (relying on expert opinion), explanation (forecasting by cause of death or using explanatory models) and extrapolation, and also by Zheng et al. (2026) who did a review on deep learning methods in mortality forecasting.

2.2 Spatio-temporal forecasting methods

Spatio-temporal methods have long been applied across a wide range of topics for forecasting and remain an active area of research, with many approaches building upon and extending established methods.

The Spatio-Temporal Autoregressive Moving Average (STARMA), established by Pfeifer and Deutrch (1980), is one that builds on a previous method, the Autoregressive Moving Average (ARMA) method. STARMA has also been extended, by researchers such as Zhao et al. (2018) who introduced the Seasonal Difference Space-Time Autoregressive Integrated Moving Average (SD-STARIMA) method, which can make seasonal difference calculations to eliminate temporally non-stationary problems that may be present in certain data, thereby improving prediction accuracy for spatio-temporal data with seasonal distribution characteristics, and by Chen et al. (2023) who proposed time-varying Spatio-Temporal Autoregressive Moving Average (tvSTARMA) modeling based on the wavelet expansion of coefficients. The STARMA method has seen a wide range of applications with Kamarianakis and Prastacos (2005) analysing traffic flow data, Kurt and Tunay (2015) regional bank deposits using Kalman Filters (KF) and Maximum Likelihood Estimation (MLE), Alam et al. (2024) rainfall, and Zou et al. (2018) wind power.

The eigenvector spatio-temporal filters (ESTF) method, introduced by Griffith (2010, 2012) also allows for the analysis of spatio-temporal data. Besides the forecasting method, he also introduces a method for calculating contemporaneous and lagged spatio-temporal autocorrelation. Extensions to ESTF have also been made with Chen et al. (2022) developing a population flow-based spatial-temporal eigenvector filtering method (FLOW-ESTF) which can take into account spatial-temporal patterns and population flow connectivity simultaneously. A review relating to the topic has been made by Griffith and Chun (2019).

The Spatial Dynamic Panel Linear Model (SDPLM) also builds on its predecessor, the Dynamic Panel Linear Model (DPLM). Key developments regarding DPLM were the introduction of a foundational method for estimating dynamic panel data models, the Anderson-Hsiao estimator, by Anderson and Hsiao (1981, 1982) and the popularisation of the Generalized Method of Moments (GMM) estimator by Arellano and Bond (1991) and the introduction of the System-GMM estimator by Blundell and Bond (1998). SDPLM

has also been extended by Yu et al. (2008) who investigated the asymptotic properties of quasi-maximum likelihood (QML) estimators for spatial dynamic panel data. Furthermore, SDPLM for both stationary and non-stationary data has been introduced by Yu and Lee (2010). A review of spatial econometrics is given by Elhorst (2014) who split it into three generations, with the first generation being the models based on cross-sectional data, the second one consisting of the non-dynamic models based on spatial panel data, and the third generation being made up of spatial dynamic panel data models.

Besides the mentioned methods, there are further variants, along with alternative approaches such as Bayesian and machine learning. Some Bayesian spatio-temporal methods can be found in these papers by Ocaña-Riola and Mayoral-Cortés (2010); Bennett et al. (2015); Njenga and Sherris (2020), and some neural network based approaches to spatio-temporal forecasting can be read about in Capone et al. (2025); Liu et al. (2019); Jiang (2020).

3 Methods

3.1 Moran's I - Spatial autocorrelation

Moran's I is the standard statistic used to quantify spatial autocorrelation. Positive values indicate that geographically close observations tend to be similar, negative values that they tend to be dissimilar, and values near zero that the pattern is close to spatial randomness (Moran, 1950).

Let N denote the total number of spatial observations, u_i $(i = 1, \ldots, N)$ an observation with $\overline{u}$ being the mean of these observations and W_{space} denote a $N \times N$ spatial weight matrix with elements w_{ij} which contain the information about the spatial relationship of observation i and j. It is common to set w_{ij} to 0 if two observations are not neighbours and set to 1 if they are. There can also be other assumptions for defining w_{ij}, with one being to assume that the strength of autocorrelation gets weaker as the distance between observations is greater in which case it is set to $\frac{1}{\gamma_{ij}}$ where γ_{ij} is the geographical distance between observation i and j which is usually measured between centroids or capital cities.

The spatial weight matrix obtained using the first method will be referred to as the adjacency spatial weight matrix, whereas the matrix derived from the second method as the inverse-distance spatial weight matrix, where distances are measured between US state capitals.

Moran's I can be formulated as the following:

$$I = \frac{N}{\sum_i \sum_j w_{ij}} \cdot \frac{\sum_i \sum_j w_{ij}(u_i - \overline{u})(u_j - \overline{u})}{\sum_i (u_i - \overline{u})^2}, \tag{1}$$

3.2 Local extension of Moran's I - LISA clustering

LISA (Local Indicators of Spatial Association) can be considered a local extension of Moran's I which breaks down Moran's I statistic into unit-level contributions and pinpoints geographic areas where neighbouring values are significantly alike forming High-High or Low–Low clusters, or especially different, forming High–Low or Low–High clusters. The local Moran's I makes new information visible which otherwise could possibly not be seen from a global spatial autocorrelation test(Anselin, 1995).

Local Moran's I using previous notation can be calculated as:

$$I_i = \frac{(u_i - \overline{u})}{\sum_k (u_k - \overline{u})^2/(N-1)} \sum_j w_{ij}(u_j - \overline{u}) \tag{2}$$

As noted by Brooks (2019), who advocates for the use of multiple LISA techniques as each one can reveal different details, there are multiple approaches to using LISA. In his paper he presents three different LISA techniques, used by him for his analysis, which are: Bivariate LISA, LISA cluster-transitions, and LISA diffusion-transitions. Bivariate LISA shows where values at a location are locally correlated with neighbours' values of a second variable. LISA cluster-transitions compares each location's local LISA cluster type at two time points and summarizes how places move between these categories. LISA diffusion-transitions detects spatial spreading or retreat of hotspots and coldspots. A further technique specifically regarding mortality forecasting was employed by Szentkereszti and Vékás (2026), who did LISA on the mean of the differenced mortality indices from their Lee–Carter and Li–Lee models.

3.3 Spatio-temporal extension of Moran's I

Spatio-temporal autocorrelation is another extension of Moran's I which characterizes the dependence of a variable across both space and time by first taking a spatial weight matrix W_{space} and expanding it via Kronecker products to make spatio-temporal weight matrix $W_{\text{space-time}}$ and then using these to formulate spatio-temporal Moran's I (Griffith, 2010).

A contemporaneous form can be written as:

$$W^{(C)}_{\text{space-time}} \;=\; I_T \otimes W_{\text{space}} \;+\; P \otimes I_N, \tag{3}$$

where P is a temporal adjacency matrix of size T that links each period to the one right before and after it, I_T is an identity matrix with size T and I_N is an identity matrix with size N.

A lagged form can be defined as:

$$W^{(L)}_{\text{space-time}} \;=\; P \otimes W_{\text{space}} \;+\; P \otimes I_N, \tag{4}$$

The contemporaneous and lagged spatio-temporal Moran's I can be written in the following form, where $\overline{u}$ denotes the overall mean over all temporal and spatial observations and $u_{(i,t)}$ the observations at location i and time t, while C and L refer to the contemporaneous and lagged specification respectively:

$$I^{(C)}_{\text{space-time}} = \frac{NT}{\sum_{(i,t)}\sum_{(j,s)} w^{(C)}_{(i,t),(j,s)}} \frac{\sum_{(i,t)}\sum_{(j,s)} w^{(C)}_{(i,t),(j,s)} \left(u_{(i,t)} - \overline{u}\right)\left(u_{(j,s)} - \overline{u}\right)}{\sum_{(i,t)} \left(u_{(i,t)} - \overline{u}\right)^2}, \tag{5}$$

$$I^{(L)}_{\text{space-time}} = \frac{NT}{\sum_{(i,t)}\sum_{(j,s)} w^{(L)}_{(i,t),(j,s)}} \frac{\sum_{(i,t)}\sum_{(j,s)} w^{(L)}_{(i,t),(j,s)} \left(u_{(i,t)} - \overline{u}\right)\left(u_{(j,s)} - \overline{u}\right)}{\sum_{(i,t)} \left(u_{(i,t)} - \overline{u}\right)^2}, \tag{6}$$

3.4 Mortality models

The modeling and projection of mortality rates will be done with the Lee–Carter (LC) and Li–Lee (LL) models, which make a stochastic approach to mortality forecasting possible. The LC model represents the logarithm of mortality rates as a combination of age specific parameters and a time-varying mortality index. Building on this framework, the LL model extends LC to a multi-population setting, allowing for coherent mortality projections across related groups by adding new parameters such as common parameters and using population-specific deviations from the common trend.

The LC model is formulated as:

$$\ln(m_{xt}) = a_x + b_x k_t + \varepsilon_{xt}, \tag{7}$$

where m_{xt} denotes the mortality rate at age x and year t and is calculated by taking the number of deaths and dividing it by the exposures at age x and year t. The age specific parameters are represented by a_x which is the average of the logarithmic mortality rates by age in the sample and b_x which denotes the sensitivity to changes in the mortality index. Furthermore, k_t denotes a time-varying mortality index and ε_{xt} is the error term. To address the issue of identification, the use of constraints is important, meaning that the sum of the k_t parameters over all t has to equal zero, while the sum of b_x over all x has to equal one. Parameters are estimated via Singular Value Decomposition (SVD) (Lee and Carter, 1992).

The LL model is formulated as:

$$\ln(m_{ixt}) = a_{ix} + B_x K_t + b_{ix} k_{it} + \varepsilon_{ixt}, \tag{8}$$

where notation follows the LC notation, expanded with common parameters B_x and K_t and a new subscript i representing a specific group, like a US state in the case of this study. To address identification, here the sum of K_t over all t has to equal the sum of k_{it} over all t and they both have to equal zero, and sum of B_x over all x has to equal the sum of b_{ix} over all x and they both have to equal one. Furthermore, to obtain coherent forecasts with

$$\lim_{t\to\infty} \frac{m_{ixt}}{m_{jxt}} = \text{constant} \qquad (i \neq j,\ x = 1, 2, \ldots, X) \tag{9}$$

all k_{it} should converge to constants over time. Parameters are estimated by repeated SVD, and K_t is forecasted with RWD estimated by MLE (Li and Lee, 2005).

3.5 Forecasting methods

Forecasting will be done on the extracted indices k_{it}, and predicted indices are mapped back into the LC and LL equations to obtain forecasts $\widehat{m}_{ixt}$.

The first method to consider is the RWD which is also what Lee and Carter (1992) used in their original paper where they introduced the LC model. Another widely used and advocated for temporal method is ARIMA which is equivalent to RWD in the special case of ARIMA(0,1,0) (Booth and Tickle, 2008; Hong et al., 2021). Both methods treat each US state's mortality index independently, assume stationarity after appropriate differencing, and are estimated by Maximum Likelihood. ARIMA orders are chosen by minimizing the Bayesian information criterion (BIC) and with a unit-root test the Augmented Dickey-Fuller (ADF) which can help us determine the necessary differencing. For further models a uniform order of differencing across states is determined via the Covariate-Augmented Dickey–Fuller (CADF) test (Costantini and Lupi, 2011). Another notable method used for mortality forecasting is the Vector Autoregression (VAR), that has both temporal and cross-sectional dependence. A key consideration here is that a large parameter count relative to the available observations can warrant the use of methods other than OLS, such as Elastic Net regularization (Zou and Hastie, 2005; Guibert et al., 2019). The number of lags in this case will be chosen by minimizing the Cross-Validated Mean Squared Error.

The next four forecasting methods were proposed by Szentkereszti and Vékás (2026), who were the first to introduce these forecasting methods into mortality forecasting and compare the performance of these methods to the ones commonly used before such as RWD, ARIMA and VAR. The estimation methods and the hyperparameter optimisation for these forecasting methods were done according to the authors' suggestions. The first novel consideration by them is the Dynamic Panel Linear Model (DPLM) that assumes a common autoregressive coefficient, in the case of this analysis for every US state, meaning it pools information across US states for estimates. The number of lags is chosen by keeping only autoregressive terms that are statistically significant, while Generalized Method of Moments (GMM), or to be exact, System-GMM is used for estimation. Moving to

spatio-temporal approaches, STARIMA by Pfeifer and Deutrch (1980) extends the previous ARIMA method by having AR and MA terms with r-th order spatial lag operators $W^{(r)}$ denoting adjacency at the r-th order and space-time lags λ_ℓ and μ_ℓ where $(\ell = 1, 2, \ldots, p)$. The number of lags and space-time lags are chosen by minimizing BIC, and the method is estimated by Kalman Filter (KF). The Spatial Dynamic Panel Linear Model (SDPLM) has separate coefficients for spatial, temporal, and space-time effects (Elhorst, 2012). They estimate the method with Quasi-Maximum Likelihood (QML) and the criterion for hyperparameter optimisation is based on the quasi-BIC, meaning the decision to include the time lag or space-time lag and whether to use the transformation proposed by Yu and Lee (2010) for bias correction when estimating with QML in addition to choosing between which spatial weight matrix to use is decided with it. It is worth highlighting that the contemporaneous term here is always included. The Eigenvector Spatio-Temporal Filter (ESTF) represents spatio-temporal structures where P is a matrix of state dummy variables and S_{space} and $S_{space-time}$ are matrices of eigenvectors of the space and space-time weight matrices (Griffith, 2012). Estimation here is going to be carried out with two different approaches: ESTF with Stepwise-OLS as originally suggested by Griffith (2003), where selection is done via BIC, and ESTF with LASSO as proposed by Seya et al. (2015) who found that LASSO can be substantially faster to compute than stepwise and that it maintains strong performance even on relatively large datasets. The selection criterion for finding the best specifications in this latter case is based on the Deviance Ratio (DR). Additionally, as a third way of estimation, following the approach of the authors, I will compute an unweighted average ensemble of the best Stepwise-OLS and LASSO specifications and compare it to the cases when either only Stepwise-OLS or LASSO is used for estimation in order to determine which estimation method performs best out of the three and should be used for estimating ESTF.

Lastly, I propose the Lagged only Spatial Dynamic Panel Linear Model (LSDPLM) estimated by System-GMM for forecasting, and the estimation of SDPLM with System-GMM as well, in addition to the previously suggested QML estimation. The use of System-GMM for dynamic panel data outside of mortality forecasting have been suggested before by authors such as Blundell and Bond (1998); Elhorst (2012). The criterion for hyperparameter optimisation will be significance similarly to the DPLM method. The LSDPLM can be viewed as a restricted version of the SDPLM, in which the contemporaneous spatial lag is excluded. This restriction makes the forecasting method more parsimonious, which could improve forecasting performance by retaining the main temporal and spatial dynamics while avoiding the additional complexity and potential instability associated with the contemporaneous term. Estimating LSDPLM with System-GMM, and SDPLM with System-GMM instead of QML, could be beneficial, considering that it provides a different, moment based approach for estimation. Furthermore, by estimating SDPLM by System-GMM and QML as well, a forecasting performance comparison is also yielded between

them in the context of this study. For additional estimation approaches regarding SDPLM, see Monteiro and Kukenova (2009).

The formulas of all the chosen methods for forecasting is summarized in Table 1 with p and q representing the number of lags, d the order of integration, T the number of calendar years, N the number of US states used in the analyses and $\xi_t = (\xi_{1t}, \xi_{2t}, \ldots, \xi_{Nt})^\top$ $(t = 1, 2, \ldots, T)$ the time-dependent error terms of the time series models

Additionally, the properties of the forecasting methods are summarized in Table 2, including the type of dependence, estimation method, hyperparameters and criterion.

Method	**Formula**
RWD	$k_{it} = \alpha_i + k_{i,t-1} + \xi_{it}$
ARIMA	$y_{it} = \Delta^{(d_i)} k_{it} = \alpha_i + \sum_{\ell=1}^{p_i} \varphi_{i\ell} y_{i,t-\ell} + \sum_{\ell=1}^{q_i} \theta_{i\ell}\, \xi_{i,t-\ell} + \xi_{it}$
DPLM	$y_t = \Delta^{(d)} k_t = \alpha + \sum_{\ell=1}^{p} \varphi_\ell y_{t-\ell} + \xi_t$
VAR	$y_t = \Delta^{(d)} k_t = \alpha + \sum_{\ell=1}^{p} A_\ell\, y_{t-\ell} + \xi_t$
STARIMA	$z_{it} = \frac{\Delta^{(d)} k_{it} - \underset{i,t}{\text{mean}}\left(\Delta^{(d)} k_{it}\right)}{\underset{i,t}{\text{sd}}\left(\Delta^{(d)} k_{it}\right)} = \alpha + \sum_{\ell=1}^{p} \sum_{r=0}^{\lambda_\ell} \phi_{\ell r}\, W^{(r)} z_{i,t-\ell} + \sum_{\ell=1}^{q} \sum_{r=0}^{\mu_\ell} \theta_{\ell r}\, W^{(r)} \xi_{i,t-\ell} + \xi_{it}$
SDPLM_QML	$y_t = \Delta^{(d)} k_t = \alpha + \phi_{\text{space}} W_{\text{space}} y_t + \phi_{\text{time}} y_{t-1} + \phi_{\text{space-time}} W_{\text{space}} y_{t-1} + \xi_t$
SDPLM_GMM	$y_t = \Delta^{(d)} k_t = \alpha + \lambda W_{space} y_t + \sum_{\ell=1}^{p} \rho_\ell y_{t-\ell} + \sum_{\ell=1}^{q} \phi_\ell W_{space} y_{t-\ell} + \xi_t$
LSDPLM_GMM	$y_t = \Delta^{(d)} k_t = \alpha + \sum_{\ell=1}^{p} \rho_\ell y_{t-\ell} + \sum_{\ell=1}^{q} \phi_\ell W_{space} y_{t-\ell} + \xi_t$
ESTF	$\Delta^{(d)} k_t = \alpha + S_{\text{space-time}} \beta + S_{\text{space}} \gamma + P\varphi + \xi_t$

Table 1: Forecasting methods
Source: Own creation based on Szentkereszti and Vékás (2026)

Method	**Dependence**	**Estimation**	**Hyperparameters**	**Criterion**
RWD	temporal	MLE	–	–
ARIMA	temporal	MLE	p_i, q_i $(i = 1, 2, \ldots, N)$	BIC
DPLM	temporal	System-GMM	p	significance
VAR	temporal and cross-sectional	ENR	p	CV-MSE
STARIMA	spatio-temporal	KF	$p,\ \lambda_\ell$ $(\ell = 1, 2, \ldots, p)$, $q,\ \mu_\ell$ $(\ell = 1, 2, \ldots, q)$	BIC
SDPLM_QML	spatio-temporal	QML	weight $\in$ {adjacency, inverse-distance}, time lag $\in \{0, 1\}$, space-time lag $\in \{0, 1\}$, Lee–Yu transf. $\in \{0, 1\}$	quasi-BIC
SDPLM_GMM	spatio-temporal	System-GMM	p, q, weight $\in$ {adjacency, inverse-distance},	significance
LSDPLM_GMM	spatio-temporal	System-GMM	p, q, weight $\in$ {adjacency, inverse-distance},	significance
ESTF	spatio-temporal	Stepwise-OLS; or LASSO	weight $\in$ {adjacency inverse-distance}, specification $\in \{C, L\}$, selection $\in$ {SW-OLS, LASSO}	BIC or DR

Table 2: Properties of used methods
Source: Own creation based on Szentkereszti and Vékás (2026)

3.6 Forecasting performance evaluation methodology

The method for performance evaluation is based on Szentkereszti and Vékás (2026), who calculated average and maximum underperformance values to compare the forecasting methods performance. The data is split into a training and test set, and as the results could be sensitive to the chosen split year, multiple train-test split years are selected to make the evaluation more robust with $T_{\text{train}}^{(h)}$ $(h = 1, 2, \ldots, H)$ representing the amount of train years and $T_{\text{test}}^{(h)}$ $(h = 1, 2, \ldots, H)$ denoting the number of test years at split h, meaning that $T = T_{\text{train}}^{(h)} + T_{\text{test}}^{(h)}$.

For forecasting method j, model ℓ, state i and split h the MSE is calculated in the following way:

$$\text{MSE}_{i,j,\ell}^{(h)} = \frac{1}{XT_{\text{test}}^{(h)}} \sum_{x=1}^{X} \sum_{t=T_{\text{train}}^{(h)}+1}^{T} \left(\ln m_{ixt} - \ln \widehat{m}_{ixt}^{(j,\ell)}\right)^2, \tag{10}$$

Underperformance relative to the best method on split h is

$$U_{i,j,\ell}^{(h)} = \text{MSE}_{i,j,\ell}^{(h)} - \min_{j,\ell} \text{MSE}_{i,j,\ell}^{(h)}, \tag{11}$$

To assess the underperformance independent of the year chosen for the train-test split an average and maximum underperformance is formulated with MU being a more conservative approach:

$$\text{AU}_{i,j,\ell} = \frac{1}{H} \sum_{h=1}^{H} U_{i,j,\ell}^{(h)}, \tag{12}$$

$$\text{MU}_{i,j,\ell} = \max_{h=1,\ldots,H} U_{i,j,\ell}^{(h)}. \tag{13}$$

4 Empirical analysis

4.1 Data sourcing and preprocessing

For the empirical analysis, I chose the United States, as mortality forecasting using SDPLM, LSDPLM, STARIMA, and ESTF has yet to be done there to the best of my knowledge. Therefore, I had to gather state by state level data on the number of deaths and exposures in the US in the past for the LC and LL models. Usually data such as this can be obtained from the Human Mortality Database (HMD, 2025), but their data for the US wasn't broken down into state by state level, so I obtained state by state data from the Harvard Dataverse Repository where Barbieri and Winant (2025) made available US life tables from 1941 to 2022 on the state, divisional, regional and national level for ages 0 to 109 and 110+ which groups together all people aged 110 or above. These life tables assumed a hypothetical starting population of 100000 people at the start of each year, therefore

I had to gather data on the actual state level exposure or death numbers to have the necessary data for the analysis. I managed to acquire real exposures data from the official US National Cancer Institute's SEER program exposure dataset [1] where exposure data for each US state was available from 1969 to 2023 for ages 0 to 89 and 90+, which groups together all the people whose age is 90 or higher. The description of life table variables can be seen in Table 3.

Variable	**Description**
$m(x)$	Mortality rate between ages x and $x+1$
$q(x)$	Probability of death between ages x and $x+1$
$a(x)$	Average number of years lived after age x by individuals who die between ages x and $x+1$.
$l(x)$	Amount of people being alive at age x
$d(x)$	Deaths between ages x and $x+1$.
$L(x)$	Person-years lived between ages x and $x+1$.
$T(x)$	Person-years remaining after age x.
$e(x)$	Life expectancy at age x.

Table 3: Life table variable descriptions
Source: Own creation based on (HMD, 2025)

I used data from 1969 to 2019 because the exposure dataset only has data available from 1969, and only until 2019 due to the Covid-19 pandemic, which could introduce unwanted bias into the analysis by making models that overestimate mortality seem better than they are. I used data between the ages 0 and 89 due to single age data only being available until 89 in the exposures dataset. I could have split the exposure data for people aged 90+ into single age groups until an even higher age around a 100 via estimation but as 89 is a high and rarely reached age, I choose to only use known single year exposures data. I also had to deal with having a very small amount (around 0.003%) of mortality rates being 0 as there were no recorded deaths in these rare cases, predominantly among younger people. Having mortality rates that are 0, in other words death values that are 0, causes a problem during mortality forecasting when log transforming the mortality rates. A common practice in literature is to add 0.5 to all of the death count values, though there are other approaches to get around this, such as using the Poisson Lee–Carter model proposed by Brouhns et al. (2002), or forecasting with P-spline smoothing established by Currie et al. (2004). I chose the common approach as a very small portion of my data is affected by this problem. Therefore, in my particular case I added 0.5 to the originally calculated death values obtained by multiplying the mortality rates with exposures and

[1] https://seer.cancer.gov/popdata/download.html#single

then recalculated the mortality rates using the modified death values. I choose the training and test data split years to be 2004, 2005, and 2006 and I have also excluded Alaska and Hawaii from the analysis, as they do not share land borders with any other US states. Although Alaska and Hawaii could be included when using an inverse-distance spatial weight matrix, they are omitted to ensure comparability between the results obtained from the two types of spatial weight matrices. Consequently, the analysis is based on the 48 states shown in Table 4.

US States excluding Alaska and Hawaii		
Alabama (AL)	Maine (ME)	Ohio (OH)
Arizona (AZ)	Maryland (MD)	Oklahoma (OK)
Arkansas (AR)	Massachusetts (MA)	Oregon (OR)
California (CA)	Michigan (MI)	Pennsylvania (PA)
Colorado (CO)	Minnesota (MN)	Rhode Island (RI)
Connecticut (CT)	Mississippi (MS)	South Carolina (SC)
Delaware (DE)	Missouri (MO)	South Dakota (SD)
Florida (FL)	Montana (MT)	Tennessee (TN)
Georgia (GA)	Nebraska (NE)	Texas (TX)
Idaho (ID)	Nevada (NV)	Utah (UT)
Illinois (IL)	New Hampshire (NH)	Vermont (VT)
Indiana (IN)	New Jersey (NJ)	Virginia (VA)
Iowa (IA)	New Mexico (NM)	Washington (WA)
Kansas (KS)	New York (NY)	West Virginia (WV)
Kentucky (KY)	North Carolina (NC)	Wisconsin (WI)
Louisiana (LA)	North Dakota (ND)	Wyoming (WY)

Table 4: US states used for the analysis
Source: Own creation

Furthermore, I will use the regional and divisional level classification of the US states by the U.S. Census Bureau (2025) throughout the analysis, which can be seen on Figure 1.

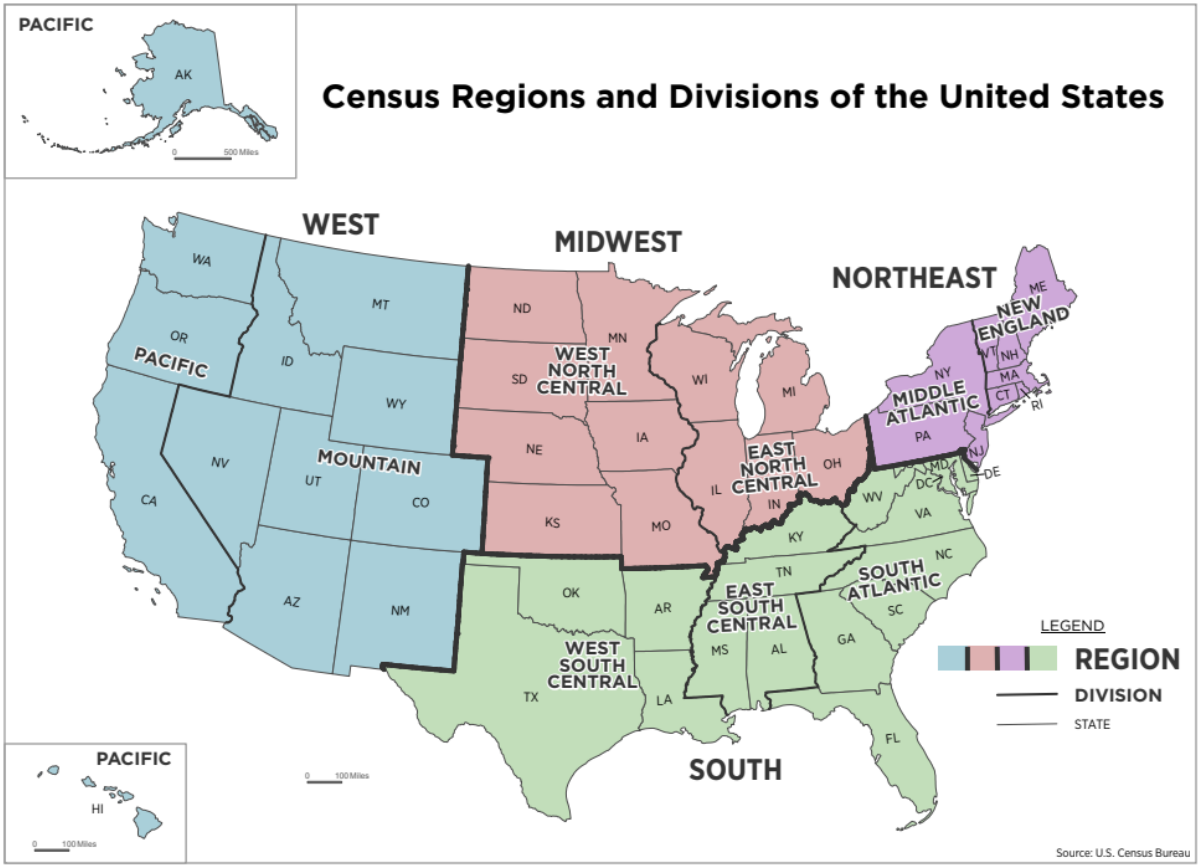

Figure 1: Census regions and divisions of the US
Source: (U.S. Census Bureau, 2025)

4.2 Analysis workflow

For the analysis, I used the R programming language [2] with libraries listed in Appendix Table A.

I first obtained the data as mentioned in the previous chapter and separated it into train and test sets with the split years being 2004, 2005, and 2006. The training data is used to estimate the time-varying mortality indices k_{it} with the LC and LL models. The LC indices are subjected to panel unit-root testing and differenced if required to achieve stationarity. Following this, each forecasting method undergoes hyperparameter optimisation and for each model type the model with best hyperparameters is selected. These forecasting models are then used with the optimised hyperparameters to forecast log-mortality rates for the test period. The performance of the forecasting methods regarding the LC and LL models are evaluated by first calculating the MSE for each forecasting method with the help of the test data for the corresponding split year, which are then used in the robust model evaluation as detailed in chapter 3.6. For a flowchart of the steps, see Figure 2.

[2]The code supporting this study's findings is available upon request

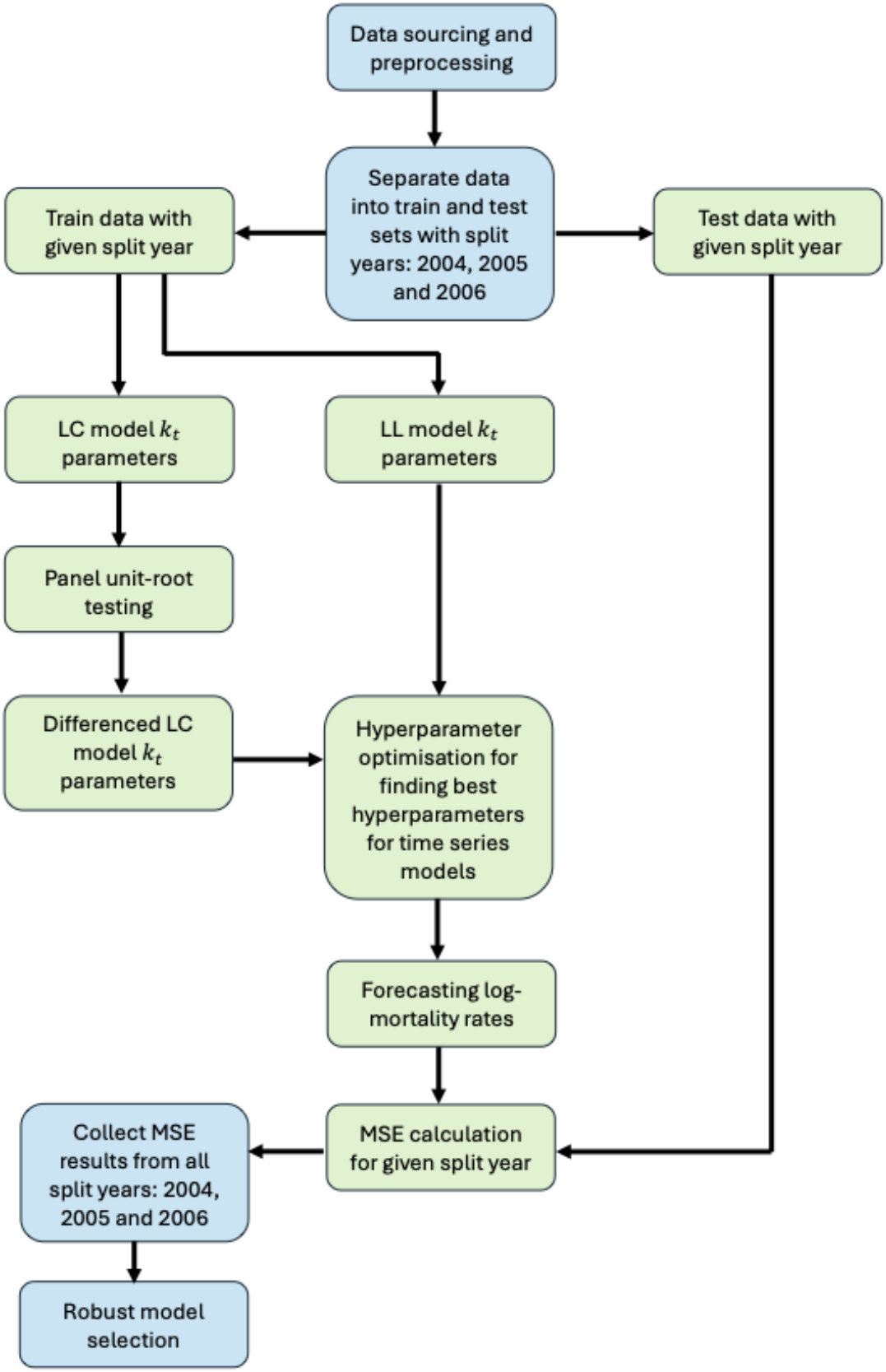

Figure 2: Flowchart of the analysis
Source: Own creation

4.3 Moran's I results for spatial and spatio-temporal autocorrelation

To justify the use of spatial forecasting methods in this analysis I calculated the spatial and spatio-temporal Moran's I statistics of the LC and LL models differenced mortality indices with the adjacency and the inverse-distance spatial weight matrix. The results can be seen in Table 5.

From these results we can see that there is a significant positive spatial autocorrelation and a significant negative contemporaneous spatio-temporal and lagged spatio-temporal autocorrelation regarding both the LC and LL models and spatial weight matrix types. Additionally, the Moran's I values regarding spatial autocorrelation when calculated with an adjacency spatial weight matrix are moderately high, whereas when calculated with an inverse-distance spatial weight matrix they are considerably lower, though still slightly positive, while in the case of contemporaneous spatio-temporal and lagged spatio-temporal autocorrelation, the difference between the values calculated with both types of weight matrix are relatively small.

Model	Spatial weight matrix type	I	$I^{(C)}_{\text{space-time}}$	$I^{(L)}_{\text{space-time}}$
LC	Adjacency	0.329	−0.068	−0.159
		(0.000)	(0.000)	(0.000)
LL	Adjacency	0.185	−0.191	−0.177
		(0.037)	(0.000)	(0.000)
LC	Inverse-distance	0.111	−0.085	−0.158
		(0.000)	(0.000)	(0.000)
LL	Inverse-distance	0.051	−0.213	−0.169
		(0.011)	(0.000)	(0.000)

Table 5: Spatial and spatio-temporal Moran's I statistics with p values in parentheses
Source: Own creation

4.4 LISA clustering results

I decided to employ two different LISA techniques in my analyses with the goal of finding out new details about the behaviour of changes in mortality in space and time regarding the US states used in the analysis and to further justify the use of spatial forecasting methods. Additionally, it is worth noting that the mortality indices used in this chapter are from the LC and LL models where the split year is chosen to be 2005 and the LISA calculations were made on the corresponding training years meaning for the years from

1969 to 2004 and using the adjacency weight matrix. [3].

First, I made an animated picture showing the LISA clustering for both the LC and LL models year over year by calculating the LISA for each year by state on the differenced mortality indices Δk_{it} at a 10% significance level. I have uploaded these animated pictures to GitHub where they can be viewed [4]. These animated pictures are useful in visually capturing short term shocks and temporal patterns regarding changes in the mortality of a given state compared to the states around it.

From the LC model's animated picture we can identify three clusters with one cluster forming from the states in the upper part of the West region and the West North Central division in the Midwest region, the second cluster forming from the South region and lower part of the Midwest region, and the third one made up of states from the Northeast region. We can see that the first cluster is a High-High cluster in the 1970s up until the middle of the 1980s and then it turns into a Low-Low cluster from towards the end of the 1980s until the end of the 1990s. In the second cluster a similar pattern can be seen but the other way around meaning it is a Low-Low cluster in 1960s, 1970s until the end of the 1980s and it turns into a High-High cluster from the beginning of the 1990s and stays like that until 2004. In the third cluster, in the 1970s and 1980s up until the middle of the 1990s the states form a High-High cluster, and then turn into a Low-Low cluster until 2004. This shift between High-High cluster to Low-Low cluster shows how in selected time frames which in this case is a few decades, certain region can turn from having worse improvement in mortality compared to the states around them into a region that has better improvement in mortality. It is interesting to note, that the clusters identified here can be divided into two separate time frames based on their cluster type and they turn into the other cluster type compared to their starting cluster type.

The LL model's animated picture shows one cluster forming from the states in the upper part of the West region and largely from the West North Central division of the Midwest region similarly to one of the LC clusters with the difference that even though it also starts out as a High-High cluster up until the end of the 1970s it changes to a Low-Low cluster in the 1980s and 1990s and then switches back to being a High-High cluster at the start of the 2000s up until 2004. A second cluster, that is a Low-Low cluster, can also be identified from 1969 up until the middle of the 1970s mainly consisting of states from the East North Central division of the Midwest region

As the second technique I did LISA clustering by taking the average of the differenced mortality indices from both LC and LL models at the previously used significance level. This LISA technique emphasizes long term changes in mortality therefore the interpretation of the results is the following here: High–High cluster means a relatively slower improvement

[3]Only the High-High, Low-Low, and Not significant cluster types are used here, as I found these to be the relevant cluster types and only using these makes interpretability clearer.

[4]https://github.com/danielk2001/TDK_spatial_mortality_forecasting

over a longer period of time in mortality compared to states around them and a Low-Low cluster means a relatively faster improvement over a longer period of time in mortality compared to states around them.

The results of the LISA clustering on the average differenced mortality indices for the LC model can be seen on Figure 3, while they can be seen for LL model on Figure 4.

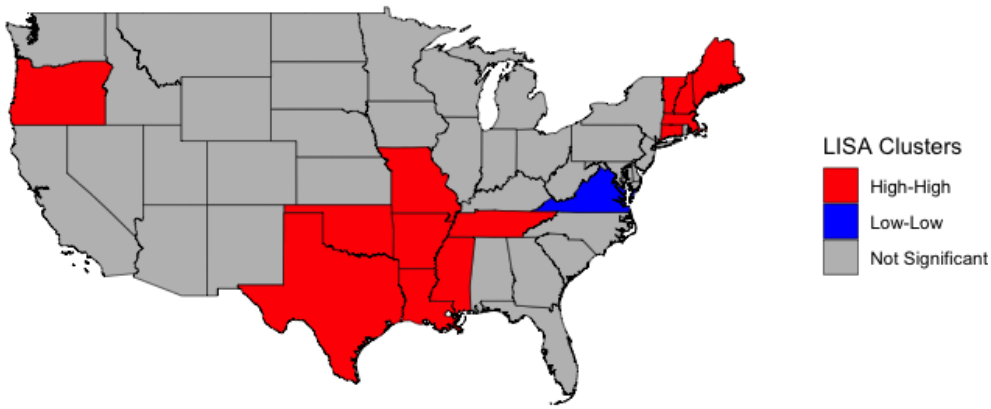

Figure 3: LISA on LC average differenced kappa
Source: Own creation

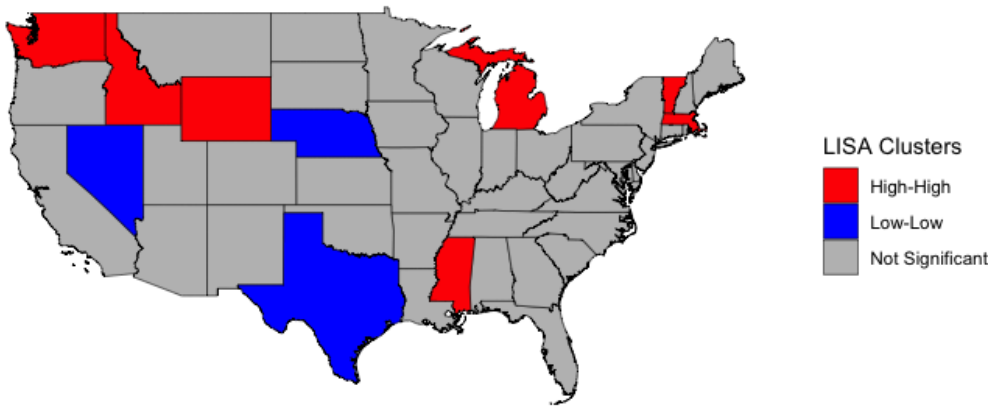

Figure 4: LISA on LL average differenced kappa
Source: Own creation

Regarding the LC model two High-High clusters can be identified. The first cluster is made up of 7 states: Texas, Oklahoma, Arkansas, Louisiana, Mississippi, Tennessee, and Missouri, meaning this cluster has predominantly states from the South region and one state from the Midwest region. The second cluster is made up of the states of Connecticut, Maine, Vermont, New Hampshire and Massachusetts, with all states being from the Northeast region.

While regarding the LL model, two smaller High-High cluster can be identified with one cluster being in the West region and the other one being in the Northeast region of the US. The first cluster consists of the state of Washington, Idaho and Wyoming and the second cluster consists of the states of Massachusetts and Vermont.

The results of the two LISA techniques show new information about the temporal patterns of changes in mortality and long term changes in mortality regarding a given state compared to the states around it and they also strengthen the case of using spatial mortality forecasting methods in this study.

4.5 Mortality forecasting

Before forecasting the log-mortality rates, it is important to optimise the hyperparameters of the chosen time series models in order to find the best hyperparameters for each one of them. The hyperparameter optimisation was done according to Table 2.

Another topic of interest before forecasting is the matter of stationarity. The stationarity of the mortality indices regarding the LC model forecasting methods except for RWD and ARIMA were tested with CADF, which gave a p value of 0.623 for the original data and a p value of 0.002 after differencing once, indicating the successful removal of unit roots, with the uniform order of integration being $d = 1$ for these methods. Concerning the LL model, as noted in a previous chapter, it is important to ensure coherence, so in this case, based on Li and Lee (2005) and Janssen (2018), the mortality indices were not differenced.

After considering the above mentioned points, mortality indices were forecasted for each state regarding both LC and LL models, using all the methods described in chapter 3.5. A visualisation of the forecasts for the ages of 40, 60, and 80, with the train-test split year being 2005, can be seen for the states of TX and NY given the forecasted mortality indices regarding the LC model on Figure 5, and the same can be seen regarding the LL model on Figure 6.

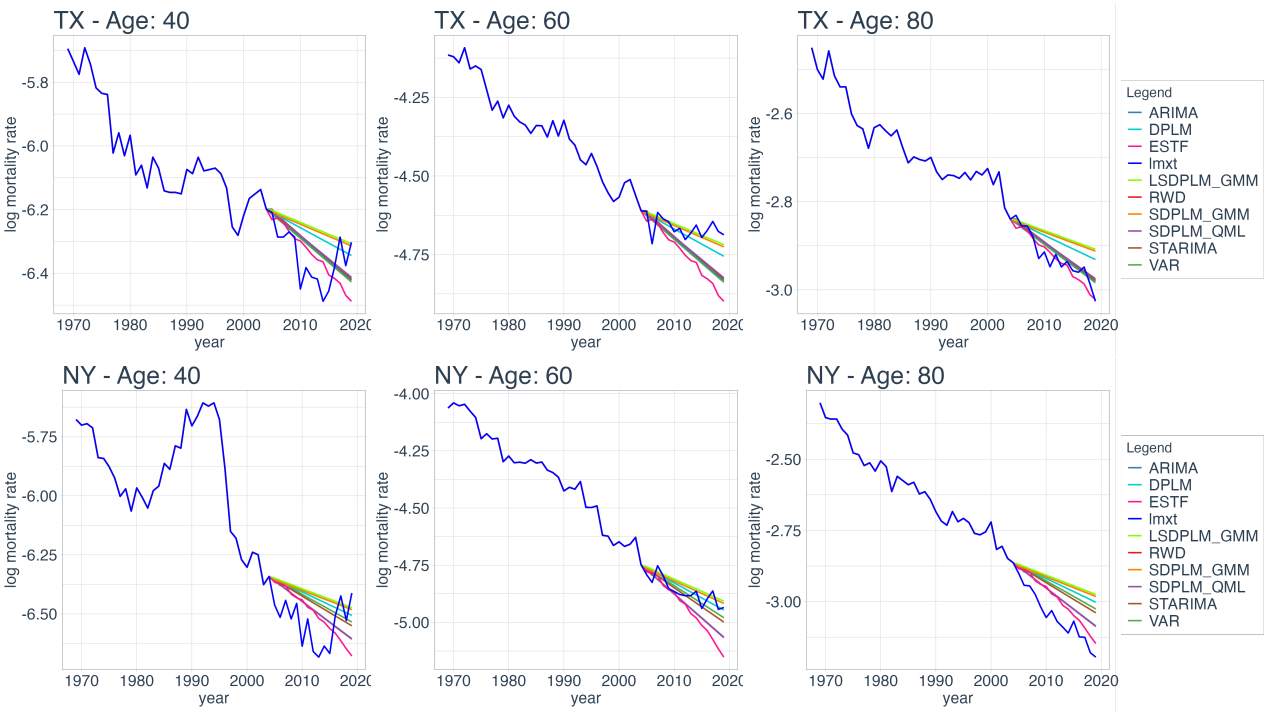

Figure 5: Forecasted log-mortality rates for certain ages and states in LC model
Source: Own creation

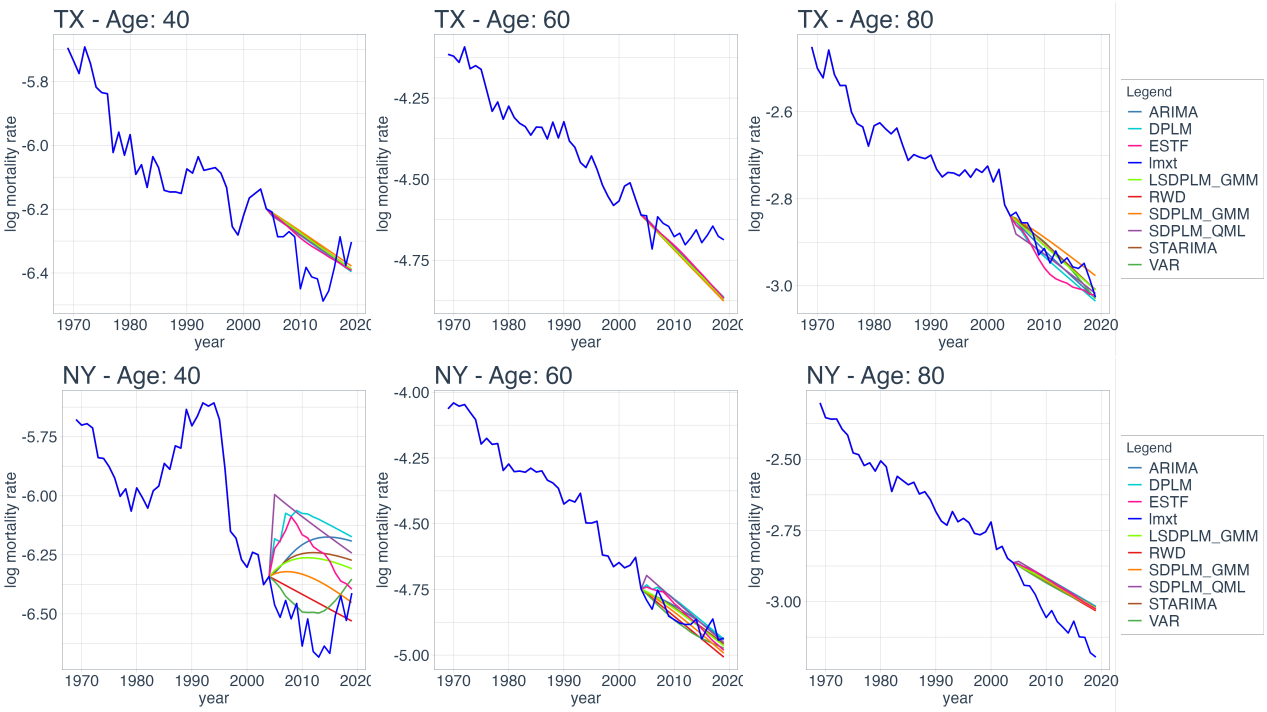

Figure 6: Forecasted log-mortality rates for certain ages and states in LL model
Source: Own creation

4.6 Model and forecasting method evaluation

Performance evaluation was done according to chapter 3.6 with the forecasted mortality indices obtained in the previous chapter considering all the chosen split years. Furthermore, I have calculated an ensemble model, which is an unweighted mean of the forecasts across all methods by state, age, and year, called Ensemble Simple Mean, and also a trimmed ensemble model, called Ensemble Trimmed Mean, which doesn't contain the highest and lowest forecasts. Following this, I have calculated the Average Underperformance (AU) and Maximum Underperformance (MU) values by state for each forecasting method and for both ensemble models regarding both the LC and LL models.

I have also calculated the Median Average Underperformance (MAU) and Median Maximum Underperformance (MMU) values to show how a single chosen method would perform if used for all states in the analysis. While using a single forecasting method is simpler and requires less computational power, if possible it is much preferable to evaluate multiple forecasting methods and choose the best one for each state. MAU and MMU values are shown on Appendix figure A.1 and A.2 respectively.

It is also worth noting that since the LC and LL models rely on fundamentally different stochastic structures and representations of mortality dynamics, their performance is evaluated separately from each other.

The results are ranked in four categories: the first category is AU values ranked from the LC model, which can be seen on Table 6, the second category is MU values ranked from the LC model shown on Table 7, and the third and fourth categories are the AU and MU values ranked from the LL model whose result can be viewed on Table 8 and Table 9, respectively. Within a ranking category, wins are achieved by each state, with the winning forecasting method for a state being the one with the lowest value by evaluation method.

Beyond comparing forecasting methods by the total number of wins, it would be worth investigating what the geographic distribution of the winning forecasting methods is, based on whether they are spatial or non-spatial method. To visualise this, I have made two maps for the LC and LL model where red coloured states represent the states where a spatial forecasting method won and the blue states represent the states where a non-spatial method won based on the more conservative MU evaluation method. The LC map is shown in Figure 7 and the LL map in Figure 8.

Following this, a visualisation of the distribution of the winning method by state could also be useful, to see whether there are any clusters where a certain forecasting method is the best performer. The visualisation for the LC model can be seen on Figure 9, while for the LL model on Figure 10 with both based on the MU evaluation method like the previous two figures.

LC Forecast results by AU

Forecasting method	Number of wins
LSDPLM_GMM	28
SDPLM_GMM	8
DPLM	5
Ensemble Simple Mean	3
Ensemble Trimmed Mean	3
SDPLM_QML	1

Table 6: LC Forecast methods in descending order by number of wins by AU
Source: Own creation

LC Forecast results by MU

Forecasting method	Number of wins
LSDPLM_GMM	23
SDPLM_GMM	14
DPLM	4
Ensemble Trimmed Mean	3
Ensemble Simple Mean	2
SDPLM_QML	1
STARIMA	1

Table 7: LC Forecast methods in descending order by number of wins by MU
Source: Own creation

LL Forecast results by AU

Forecasting method	Number of wins
SDPLM_GMM	14
RWD	10
VAR	10
ARIMA	7
Ensemble Simple Mean	3
DPLM	1
ESTF	1
SDPLM_QML	1
STARIMA	1

Table 8: LL Forecast methods in descending order by number of wins by AU
Source: Own creation

LL Forecast results by MU

Forecasting method	Number of wins
SDPLM_GMM	14
RWD	10
VAR	9
ARIMA	7
Ensemble Simple Mean	3
DPLM	2
ESTF	2
STARIMA	1

Table 9: LL Forecast methods in descending order by number of wins by MU
Source: Own creation

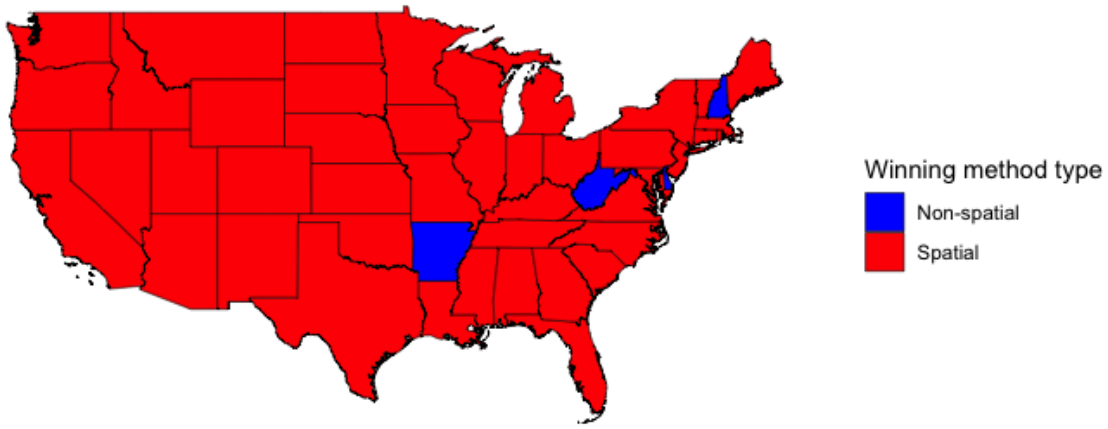

Figure 7: Distribution of spatial and non-spatial LC winner methods by MU
Source: Own creation

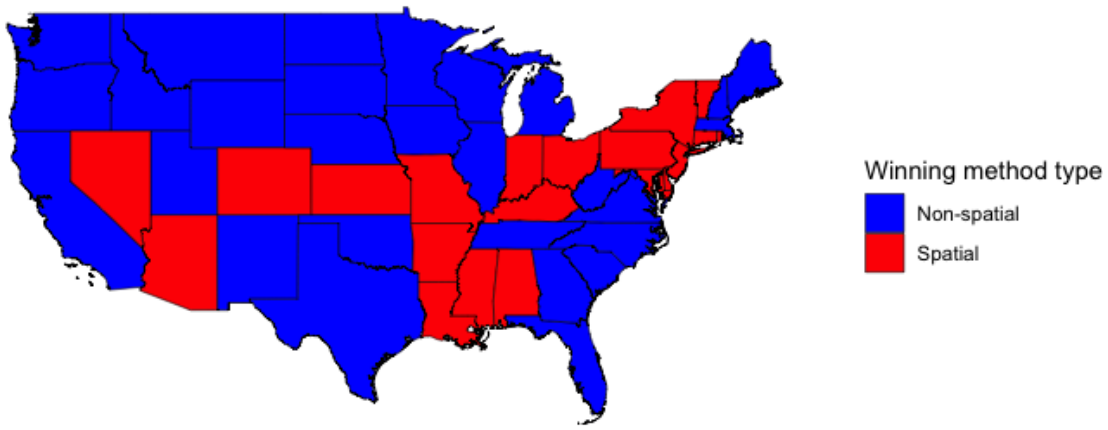

Figure 8: Distribution of spatial and non-spatial LL winner methods by MU
Source: Own creation

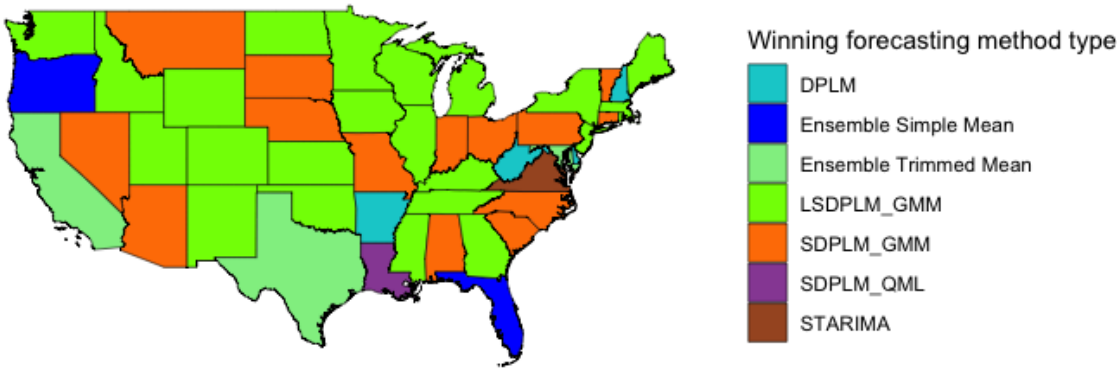

Figure 9: Distribution of the winning method by state in the LC model by MU
Source: Own creation

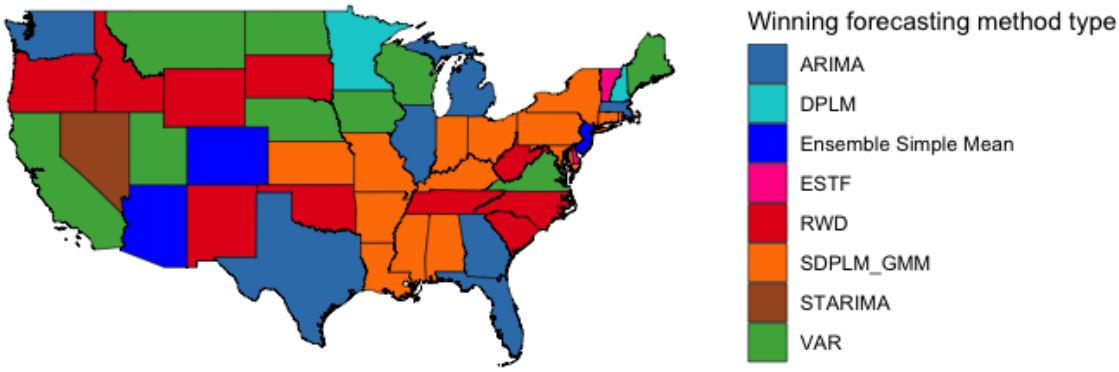

Figure 10: Distribution of the winning method by state in the LL model by MU
Source: Own creation

Based on these results, both AU and the more conservative MU, as shown in the tables above, produced similar results within the LC and LL models respectively. In the case of the LC model, spatial models clearly outperformed non-spatial ones in the US under the evaluation method used in this study, with the only non-spatial winning method being the DPLM, which won in 5 states by AU and 4 times by MU. The best performing method in the context of the study was the proposed LSDPLM estimated by System-GMM, which performed considerably better than other methods, which may be due to it striking the right balance between incorporating spatiality into mortality forecasting and method complexity. The LSDPLM in LC won in around 58% of the states by AU and 48% of the states by MU. The other proposed forecasting method, the SDPLM estimated by System-GMM also performed well and took the second place by both AU and MU.

This may suggest that both spatial and non-spatial dynamic panel mortality forecasts estimated by System-GMM are well suited for mortality forecasting in the US and potentially elsewhere as well. The success of the dynamic panel based methods could possibly be due to the choice of estimation method, as the System-GMM version of SDPLM performed better in the analysis than the one estimated by QML. It might also stem from a relatively higher degree of homogeneity of the states across the US in many aspects (or at least homogeneous enough for methods that pool together the data from all the states and assume common autoregressive coefficients in their method to potentially perform well), despite individual states having variations in laws, policies, cultural traditions and other socioeconomic factors. In other words, one possible explanation is that due to the dynamic panel based methods estimating common coefficients in their formula and the data being relatively homogeneous, the method can capture information well, while also being more parsimonious than some other methods and also having the option to take spatiality into account, which may further increase performance.

In the case of the LL model a slightly different picture can be seen with non-spatial methods winning in around 58% of the states by both AU and MU, though the best performing method in the analysis, the SDPLM estimated by System-GMM, is still a spatial and dynamic panel based method. This shift towards non-spatial methods performing better could be due to the LL method capturing a sizeable amount of the cross state information with its common parameters, and therefore making spatial models perform worse, as their spatial terms are left with less information to capture, while increasing method complexity. It is also interesting to note, that the second, third and fourth best performing methods were the RWD, VAR, and ARIMA respectively, which are the classic forecasting methods used for mortality forecasting. Additionally, spatial methods such as ESTF, STARIMA, and SDPLM estimated by QML also managed to get wins, with the ensemble methods also obtaining some wins in both the LC and LL case.

Another topic of interest, is the clusters formed by states based on which type of forecasting method won. In the context of the empirical analysis, when only taking

into account whether a method is spatial or not, regarding the LC model, see Figure 7, non-spatial clusters cannot be identified, as in almost all the states spatial methods came out as winners. On the other hand, in the LL cases, see Figure 8, multiple spatial and non-spatial clusters can be seen. One notable spatial cluster in this case is the cluster forming from states that are located in the East North Central division of the Midwest region and the Middle Atlantic division of the Northeast region, with another notable cluster being made up of states from the central part of the South region and the southern part of the Midwest.

Furthermore, if taken into account what particular method won by each state and not only whether they are spatial or not, then regarding the LC model, see Figure 9, there are two large spatial forecasting method based clusters. The first large cluster is formed by states from the Western region, where LSDPLM won, and the second large cluster is made up of states from the Midwest, where LSDPLM is also the winning method. Other notable clusters in the analysis are also formed across the US in which the best forecasting method seems to be SDPLM estimated by System-GMM. The other methods that won in this case don't seem to form clusters. This may suggest, based on the observed clusters when using the LC model, that in the two larger areas lagged spatial dependence is more relevant than contemporaneous, while in a few smaller areas with the SDPLM clusters, the inclusion of contemporaneous spatial dependence adds predictive value. When it comes to the LL model, see Figure 10, a few smaller clusters can be seen where RWD and ARIMA won, and two larger sub-clusters of spatial method based clusters can be seen where SDPLM estimated by System-GMM won, making up almost the whole spatial cluster as seen on Figure 8. These findings appear to imply that the incorporation of spatiality into mortality forecasting, the use of the newly proposed LSDPLM, and the estimation of LSDPLM and SDPLM by System-GMM are worthwhile.

The comparison of these results with previous findings is difficult, as the literature on the topic of mortality forecasting with dynamic panel based methods, STARIMA and ESTF is very limited, with Szentkereszti and Vékás (2026) being one who performed an analysis on 22 European countries using spatio-temporal methods such as SDPLM, STARIMA and ESTF. In their empirical analysis, in the LC case, their best method was the ESTF, with the SDPLM estimated by QML also performing well under both AU and MU. In the LL case, by both evaluation methods, DPLM was their best method, with RWD and VAR also performing reasonably well. Additionally, spatio-temporal methods also performed well regarding the LL model, with STARIMA emerging as a strong performer. When it comes to the type of spatial weight matrix used, in their analysis the adjacency weight matrix clearly outperformed the inverse-distance weight matrix, whereas in this study there was no clear spatial weight matrix type that outperformed the other one in most cases. Based on their findings, it seems that even on data on European countries which arguably have lower homogeneity, the dynamic panel based models still performed strongly, further

strengthening the argument for taking them into account in mortality forecasting. The great performance of spatial mortality forecasting methods in their paper, this study, and other researches using Bayesian and machine-learning based spatio-temporal forecasting methods further support the idea that the consideration of spatiality when doing mortality forecasting could be important.

5 Conclusion

In this study, the performance of temporal and spatio-temporal mortality forecasting methods within the Lee–Carter and Li–Lee frameworks is compared using United States mortality data and multiple train–test splits for robust evaluation. Additionally, two types of LISA clustering techniques are employed to provide insights into temporal patterns of mortality change and long-term mortality change regarding a given state compared to the states around it. Figures are also presented showing the winning method by state, as well as whether the winning method is spatial or non-spatial.

The main contribution of this study is the empirical comparison of traditional temporal and spatio-temporal mortality forecasting methods within the Lee–Carter and Li–Lee frameworks using United States mortality data. In addition, the study introduces the LSDPLM method (estimated by System-GMM), estimates the SDPLM with System-GMM alongside previously used approaches in this context, applies LISA clustering as a complementary tool for spatial interpretation of mortality patterns in the United States, and presents figures showing the spatial distribution of the winning forecasting methods by state, thereby allowing for further examination of spatial patterns and identification of clusters based on which forecasting method won in each state.

In the case of the LC model, the best forecasting method in the context of this study was the proposed LSDPLM estimated by System-GMM, with the SDPLM estimated by System-GMM and DPLM also showing strong performances. In this instance, spatial methods outperformed non-spatial approaches in almost all states included in the analysis, with every identified cluster across the US consisting of spatial forecasting methods, resulting in two larger clusters where the LSDPLM won, and some other notable clusters in which the SDPLM estimated by System-GMM performed the best.

In the case of the LL model, the best performing mortality forecasting method in the context of the study was the SDPLM estimated by the suggested System-GMM, with traditional methods such as RWD, ARIMA and VAR also achieving great results. Furthermore, both spatial and non-spatial forecasting method based cluster are present, with a few smaller clusters where RWD and ARIMA achieved the best results, and two larger clusters in which SDPLM estimated by System-GMM was the best performer.

Nevertheless, the study is subject to a number of constraints. It is restricted to the LC and LL models, with the LL common parameter K_t being forecasted only by RWD,

and it does not incorporate Bayesian or machine learning based approaches. Additionally, alternative weight matrix types could also be explored besides the adjacency and inverse-distance weight matrices. Other methods could also be applied to handle zero mortality rates and more than three train–test splits could also be used. The study considers only observed single ages between 0 and 89 and could be advanced by not only having point but also interval forecasts, which could be constructed following the approach outlined by Szentkereszti and Vékás (2026). In the future, it would also be valuable to apply all the methods used in this comparative study to other regions, countries, or even more granular data to see how the results may change.

Advancements in the area of mortality forecasting can provide valuable practical benefits for actuaries and demographers by enabling more accurate mortality forecasting and demographic projections, thus supporting improved decision making in areas such as pension, life insurance, and public policymaking.

References

Ahmadi, S. S. and Li, J. S.-H. (2014). Coherent mortality forecasting with generalized linear models: A modified time-transformation approach. *Insurance: Mathematics and Economics*, 59:194–221.

Alam, N. M., Mitra, S., Pandey, S. K., Jana, C., Ray, M., Ghosh, S., Paul Mazumdar, S., Shankar, S. V., Saha, R., and Kar, G. (2024). Enhanced spatio-temporal modeling for rainfall forecasting: A high-resolution grid analysis. *Water*, 16(13):1891.

Anderson, T. W. and Hsiao, C. (1981). Estimation of dynamic models with error components. *Journal of the American Statistical Association*, 76(375):598–606.

Anderson, T. W. and Hsiao, C. (1982). Formulation and estimation of dynamic models using panel data. *Journal of Econometrics*, 18(1):47–82.

Anselin, L. (1995). Local indicators of spatial association—LISA. *Geographical analysis*, 27(2):93–115.

Arellano, M. and Bond, S. (1991). Some tests of specification for panel data: Monte carlo evidence and an application to employment equations. *Review of Economic Studies*, 58(2):277–297.

Barbieri, M. and Winant, C. (2025). U.S State Life Tables CSV + TXT 1941-2022. DOI: https://doi.org/10.7910/DVN/19WYUX.

Barrieu, P., Bensusan, H., El Karoui, N., Hillairet, C., Loisel, S., Ravanelli, C., and Salhi, Y. (2012). Understanding, modelling and managing longevity risk: Key issues and main challenges. *Scandinavian Actuarial Journal*, 2012(3):203–231.

Bennett, J. E., Li, G., Foreman, K., Best, N., Kontis, V., Pearson-Stuttard, J., and Ezzati, M. (2015). The future of life expectancy and life expectancy inequalities in England and Wales: Bayesian spatiotemporal forecasting. *The Lancet*, 386(9989):163–170.

Blundell, R. and Bond, S. (1998). Initial conditions and moment restrictions in dynamic panel data models. *Journal of Econometrics*, 87(1):115–143.

Boonen, T. J. and Li, H. (2017). Modeling and forecasting mortality with economic growth: A multipopulation approach. *Demography*, 54(5):1921–1946.

Booth, H. and Tickle, L. (2008). Mortality modelling and forecasting: A review of methods. *Annals of Actuarial Science*, 3(1-2):3–43.

Brooks, M. M. (2019). The Advantages of Comparative LISA Techniques in Spatial Inequality Research: Evidence from Poverty Change in the United States. *Spatial Demography*, 7:167–193.

Brouhns, N., Denuit, M., and Vermunt, J. K. (2002). A poisson log-bilinear regression approach to the construction of projected lifetables. *Insurance: Mathematics and Economics*, 31(3):373–393.

Cairns, A. J. G., Blake, D., and Dowd, K. (2006). A two-factor model for stochastic mortality with parameter uncertainty: Theory and calibration. *Journal of Risk and Insurance*, 73(4):687–718.

Cairns, A. J. G., Blake, D., Dowd, K., Coughlan, G. D., Epstein, D., and Khalaf-Allah, M. (2011). Mortality density forecasts: An analysis of six stochastic mortality models. *Insurance: Mathematics and Economics*, 48(3):355–367.

Cairns, A. J. G., Blake, D., Dowd, K., Coughlan, G. D., Epstein, D., Ong, A., and Balevich, I. (2009). A quantitative comparison of stochastic mortality models using data from England & Wales and the United States. *North American Actuarial Journal*, 13(1):1–35.

Capone, V., Casolaro, A., and Camastra, F. (2025). Spatio-temporal prediction using graph neural networks: A survey. *Neurocomputing*, 643:130400.

Chen, M., Chen, Y., Xu, Y., An, Q., and Min, W. (2022). Population flow based spatial-temporal eigenvector filtering modeling for exploring effects of health risk factors on covid-19. *Sustainable Cities and Society*, 87:104256.

Chen, Y., Morettin, P. A., and Chiann, C. (2023). Time-varying STARMA models by wavelets. *arXiv preprint arXiv:2304.06110*.

Costantini, M. and Lupi, C. (2011). A Simple Panel-CADF Test for Unit Roots. Technical Report 261, Institute for Advanced Studies (IHS), Vienna.

Currie, I. D., Durban, M., and Eilers, P. H. C. (2004). Smoothing and forecasting mortality rates. *Statistical Modelling*, 4(4):279–298.

Dowd, K., Cairns, A. J. G., Blake, D., Coughlan, G. D., and Khalaf-Allah, M. (2011). A gravity model of mortality rates for two related populations. *North American Actuarial Journal*, 15(2):334–356.

Elhorst, J. P. (2012). Dynamic spatial panels: Models, methods, and inferences. *Journal of Geographical Systems*, 14(1):5–28.

Elhorst, J. P. (2014). *Spatial Econometrics: From Cross-Sectional Data to Spatial Panels*. Springer.

Enchev, V., Kleinow, T., and Cairns, A. J. G. (2017). Multi-population mortality models: fitting, forecasting and comparisons. *Scandinavian Actuarial Journal*, 2017(4):319–342.

Gompertz, B. (1825). On the nature of the function expressive of the law of human mortality, and on a new mode of determining the value of life contingencies. *Philosophical Transactions of the Royal Society of London*, 115:513–585.

Griffith, D. A. (2003). *Spatial Autocorrelation and Spatial Filtering: Gaining Understanding Through Theory and Scientific Visualization*. Advances in Spatial Science. Springer, Berlin, Heidelberg.

Griffith, D. A. (2010). Modeling spatio-temporal relationships: retrospect and prospect. *Journal of Geographical Systems*, 12:111–123.

Griffith, D. A. (2012). Space, time, and space-time eigenvector filter specifications that account for autocorrelation. *Estadística Española*, 54(177):7–34.

Griffith, D. A. and Chun, Y. (2019). *Spatial Regression Analysis Using Eigenvector Spatial Filtering*. Elsevier, Amsterdam.

Guibert, Q., Loisel, S., Lopez, O., and Piette, P. (2020). Bridging the Li-Carter's Gap: A Locally Coherent Mortality Forecast Approach. Working Paper hal-02472777, HAL Open Science.

Guibert, Q., Lopez, O., and Piette, P. (2019). Forecasting mortality rate improvements with a high-dimensional VAR. *Insurance: Mathematics and Economics*, 88:255–272.

HMD (2025). Explanatory notes. https://www.mortality.org/Data/ExplanatoryNotes.

Hong, W.-H., Yap, J.-H., Selvachandran, G., Thong, P. H., and Son, L. H. (2021). Forecasting mortality rates using hybrid Lee–Carter model, artificial neural network and random forest. *Complex Intell. Syst.*, 7(1):163–189.

Hunt, A. and Blake, D. (2014). A general procedure for constructing mortality models. *North American Actuarial Journal*, 18(1):116–138.

Hunt, A. and Villegas, A. M. (2015). Robustness and convergence in the Lee–Carter model with cohort effects. *Insurance: Mathematics and Economics*, 64:186–202.

Janssen, F. (2018). Advances in mortality forecasting: Introduction. *European Journal of Epidemiology*, 33(9):801–808.

Jarner, S. F. and Kryger, E. M. (2011). Modelling adult mortality in small populations: The saint model. *ASTIN Bulletin*, 41(2):377–418.

Jiang, Z. (2020). A survey on spatial and spatiotemporal prediction methods. https://arxiv.org/abs/2012.13384.

Kamarianakis, Y. and Prastacos, P. (2005). Space–time modeling of traffic flow. *Computers & Geosciences*, 31(2):119–133.

Kurt, S. and Tunay, K. B. (2015). STARMA Models Estimation with Kalman Filter: The Case of Regional Bank Deposits. *Procedia - Social and Behavioral Sciences*, 195:2537–2547.

Lee, R. D. and Carter, L. R. (1992). Modeling and Forecasting U.S. Mortality. *Journal of the American Statistical Association*, 87(419):659–671.

Levantesi, S. and Pizzorusso, V. (2019). Application of machine learning to mortality modeling and forecasting. *Risks*.

Li, N. and Lee, R. D. (2005). Coherent mortality forecasts for a group of populations: An extension of the Lee–Carter method. *Demography*, 42(3):575–594.

Liu, M., Ke, Y., Yin, Q., Chen, X., and Im, J. (2019). Comparison of five spatio-temporal satellite image fusion models over landscapes with various spatial heterogeneity and temporal variation. *Remote Sensing*, 11(22).

Macdonald, A., Gallop, A., Miller, K., Richards, S., Shah, R., and Willets, R. (2007). Stochastic projection methodologies: Lee–Carter model features, example results and implications. Working paper 25 (revised), Continuous Mortality Investigation (Life Office Mortality Committee).

Monteiro, J.-A. and Kukenova, M. (2009). Spatial Dynamic Panel Model and System GMM: A Monte Carlo Investigation. IRENE Working Papers 09-01, IRENE Institute of Economic Research.

Moran, P. A. P. (1950). A test for the serial independence of residuals. *Biometrika*, 37(1–2):178–181.

Njenga, C. N. and Sherris, M. (2020). Modeling mortality with a Bayesian vector autoregression. *Insurance: Mathematics and Economics*, 94:40–57.

Ocaña-Riola, R. and Mayoral-Cortés, J. M. (2010). Spatio-temporal trends of mortality in small areas of Southern Spain. *BMC Public Health*, 10:26.

Pfeifer, P. E. and Deutrch, S. J. (1980). A three-stage iterative procedure for space-time modeling phillip. *Technometrics*, 22(1):35–47.

Plat, R. (2009). On stochastic mortality modeling. *Insurance: Mathematics and Economics*, 45(3):393–404.

Renshaw, A. E. and Haberman, S. (2006). A cohort-based extension to the Lee–Carter model for mortality reduction factors. *Insurance: Mathematics and Economics*, 38(3):556–570.

Richman, R. and Wüthrich, M. V. (2021). A neural network extension of the Lee–Carter model to multiple populations. *Annals of Actuarial Science*, 15(2):346–366.

Seya, H., Murakami, D., Tsutsumi, M., and Yamagata, Y. (2015). Application of lasso to the eigenvector selection problem in eigenvector-based spatial filtering. *Geographical Analysis*, 47(3):284–299.

Shen, Y., Yang, X., Liu, H., et al. (2024). Advancing mortality rate prediction in european population clusters: integrating deep learning and multiscale analysis. *Scientific Reports*, 14:6255.

Szentkereszti, G. and Vékás, P. (2026). Mortality Forecasting in Geographical Space and Time. *ASTIN Bulletin*, pages 1–22.

U.S. Census Bureau (2025). Reference maps. `https://www2.census.gov/geo/pdfs/maps-data/maps/reference/us_regdiv.pdf`.

Wan, C. and Bertschi, L. (2015). Swiss coherent mortality model as a basis for developing longevity de-risking solutions for Swiss pension funds: A practical approach. *Insurance: Mathematics and Economics*, 63:66–75.

Yu, J., de Jong, R., and Lee, L.-F. (2008). Quasi-maximum likelihood estimators for spatial dynamic panel data with fixed effects when both n and t are large. *Journal of Econometrics*, 146(1):118–134.

Yu, J. and Lee, L.-F. (2010). Estimation of unit root spatial dynamic panel data models. *Econometric Theory*, 26(5):1332–1362.

Zhao, Y., Ge, L., Zhou, Y., Sun, Z., Zheng, E., Wang, X., et al. (2018). A new Seasonal Difference Space-Time Autoregressive Integrated Moving Average (SD-STARIMA) model and spatiotemporal trend prediction analysis for Hemorrhagic Fever with Renal Syndrome (HFRS). *PLoS ONE*, 13(11):e0207518.

Zheng, H., Wang, H., Zhu, R., and Xue, J.-H. (2026). A brief review of deep learning methods in mortality forecasting. *Annals of Actuarial Science*, 20(1):150–165.

Zou, H. and Hastie, T. (2005). Regularization and variable selection via the elastic net. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 67(2):301–320.

Zou, J., Zhu, J., Xie, P., Xuan, P., and Lai, X. (2018). A STARMA Model for Wind Power Space-Time Series. In *2018 IEEE Power & Energy Society General Meeting (PESGM)*, pages 1–5. IEEE.

Appendix

Package name
bigstep
CADFtest
cepiigeodist
cowplot
data.table
demography
dplyr
fastDummies
forecast
fUnitRoots
ggplot2
ggthemes
glmnet
gridExtra
lmtest
MASS
Matrix
MultiMoMo
patchwork
plm
punitroots
readxl
remotes
rnaturalearth
rnaturalearthdata
rnaturalearthhires
sf
sparsevar
spdep
spatialreg
starma
StMoMo
tidyr
tidyquant
timetk
tseries

Table A: R packages used in the analysis

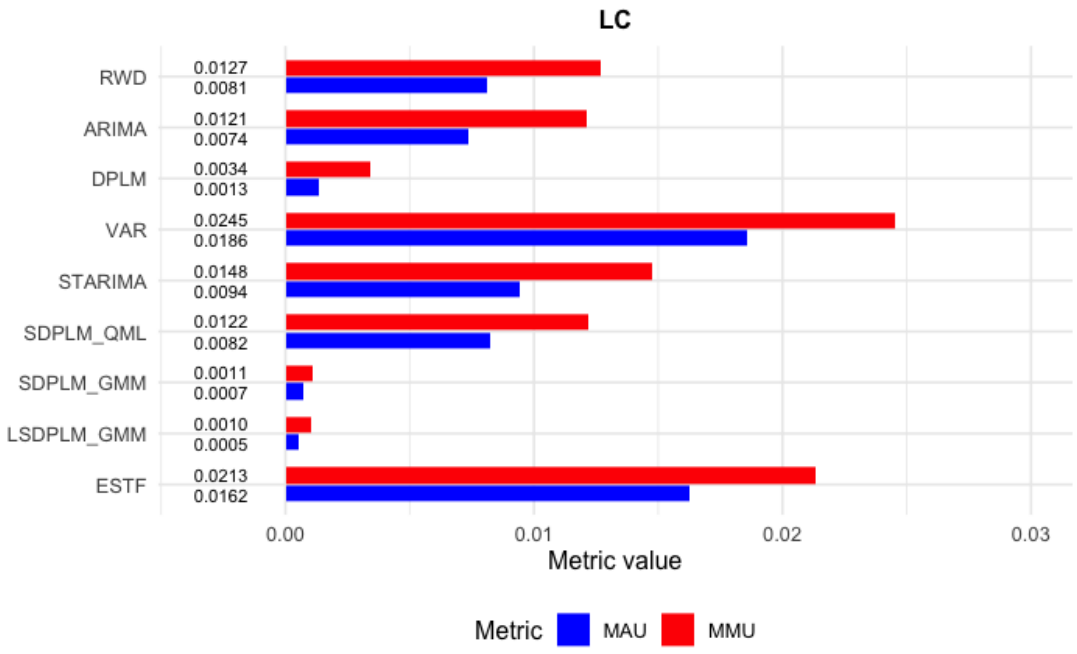

Figure A.1: MAU and MMU values by forecasting method regarding LC
Source: Own creation

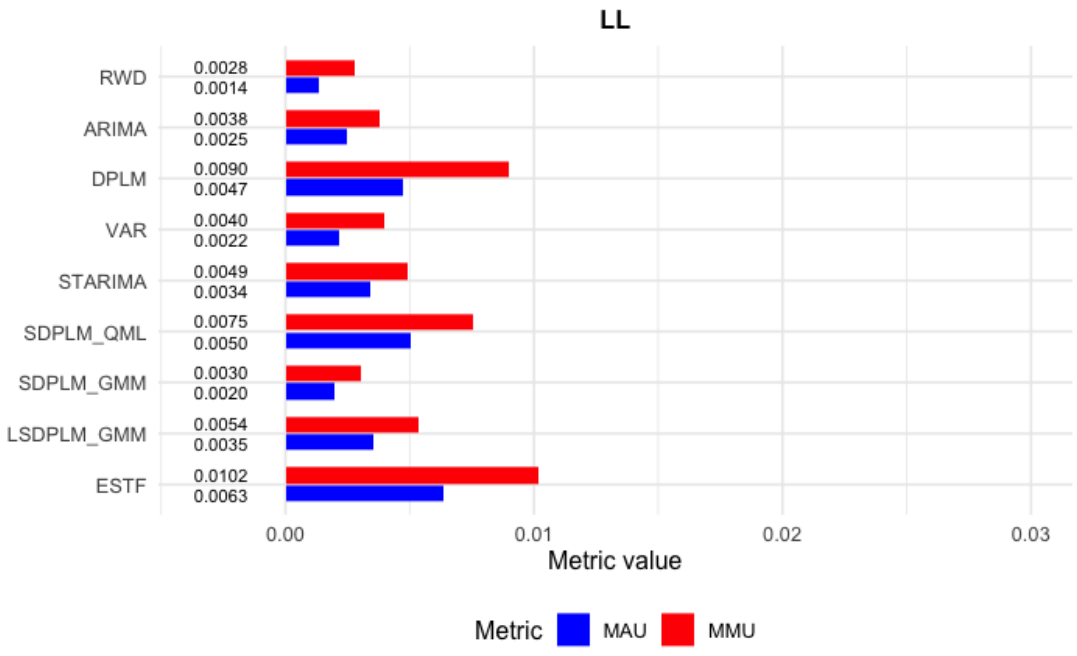

Figure A.2: MAU and MMU values by forecasting method regarding LL
Source: Own creation